

EMG Sensor Signal Processing and Communication With a Serious Game for Stroke Rehabilitation

1st Gabriel Rael da Silva

Postgraduate Program in Mechanical Engineering

Federal University of Santa Maria (UFSM)

Santa Maria, Brazil

g.rael@hotmail.com

Abstract—This work deals with the acquisition of electromyography (EMG) signals through a generic muscle sensor, the signal's processing and treatment so it can be better used in a serious game for rehabilitation made in the Unity engine. After research of related works, the author implements some signal filtering algorithms to the received EMG signal: a median filter, an adaptive low-pass filter and an exponential filter. All of them show some potential, specially the exponential filter, which suggests more experimentation with this method.

Index Terms—EMG, Unity, Microcontrollers, Serious Game, Signal Processing

I. INTRODUCTION

Physical therapists have used serious EMG-based games for the rehabilitation of patients with poststroke rehabilitation for many years [1], improving the effects of treatment when combined with traditional methods and exercises. In a previous work [2] the author made use of a generic low-cost EMG sensor to capture the contractions of the muscles responsible for opening the hand for a serious game designed for the rehabilitation of patients with post-stroke hemiparesis.

The EMG signal was processed in a ESP32 microcontroller and transmitted to a serious game made into the Unity engine via Bluetooth, to be used as input for the character's movement. Although the signal passes through initial processing on the integrated circuit of the sensor [3], it can still provide noisy output, and transmission may be limited by the speed of Bluetooth communication and connectivity stability. Also, as the game also runs on a limited framerate, running the code only a few dozens of times per second, only a small number of the readings can be received by the software, even if the sensor and microcontroller can process thousands of them each second.

These limitations lead to a software based solution where a moving average of sensor readings would be calculated and sent to the game with a interval of 100 milliseconds. While this approach did result in satisfying game response and without any noticeable delays, it is expected that a more refined method of signal processing, filtering, and sampling could achieve greater results. The goal of this work is to develop one such method.

II. RELATED WORKS

Other works have used EMG sensors similar to the one used in this project, making use of different filtering technologies

as needed for each application.

The work of [4] makes use of a newer sensor from Advancer technologies, the MyoWare, investigating the accuracy of its readings and how to properly analyze the data gathered. Their recorded surface EMG (sEMG) signal was in the range of 400-500Hz, so a sampling frequency of 2000Hz was considered sufficient as per the Nyquist rate. As sEMG is usually accompanied by noise from cross talk, motion artifacts and ambient noise, the researchers implemented a Butterworth filter to pass typical EMG frequencies (4-500Hz) and a band stop filter to eliminate power line frequency of 50Hz (which would be 60Hz here).

Other researchers [5] designed their own EMG sensors to be used in a virtual reality (VR) game for rehabilitation. Although their sensor differs from the one used here, it is integrated to the same game engine, Unity. As the communication between microcontroller and the engine is asynchronous in their project, the absolute value of 64 samples is taken and averaged, so to avoid sub-sampling errors. This project utilizes synchronous communication, so readings were not lost, but if they arrived faster than the engine's update rate, then they would pile up and cause delays, so a similar averaging approach and sampling would be equally useful. The authors also implement a threshold in the game engine so that only values above it are considered to modify the game's animation, a similar approach was used in this work.

Another work [6] uses the same model of sensor as this one (Muscle Sensor V3). These authors also average the EMG signal from windows of 24 samples to reduce noise. Median or average filters are usual for reducing noise in EMG signals, as seen in other works [7] [8] [9].

III. THE SERIOUS GAME

As cited in the Introduction, the author developed previously a serious game based on EMG signals to be used in rehabilitation of post-stroke patients [2]. The goal of this game was the training of the muscles responsible for opening the hand (extensor digitorum), as post-stroke patients with hemiparesis or hemiplegia often suffer from a permanently clenched fist [10].

This game consisted of a screen showing a fish character moving by itself from left to right, as shown in Figure 1, but that would move upwards when the user contracted the muscle

being trained and go down when the muscle was relaxed. To gather points, the player has to control the fish to collect as many coins as possible, having a total score and a precision percentage shown to them during the game and on the end screen.

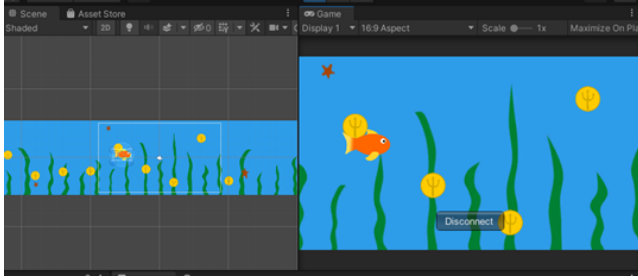


Fig. 1. The game screen as captured inside the development environment.

A virtual hand was also shown on a second screen, to be placed in front of the patient arm, as seen on Figure 2. This hand would open as the muscle was contracted and close when relaxed, even if the real hand couldn't do so. This was done to emulate mirror therapy.

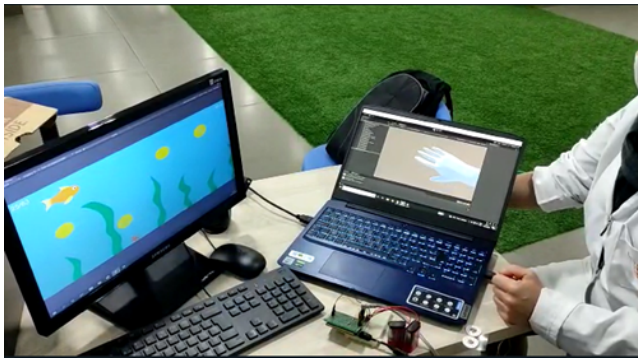


Fig. 2. The serious game prototype being tested by a physiotherapist.

A new version is now under development, using a VR headset instead of two monitors. The game screen with the fish is now displayed on a computer screen inside a virtual reality environment, while the hand is replaced by the one on the user's VR avatar, as it can be seen on Figure 3

IV. EMG SENSOR

In order to have a muscle contraction as input for the serious game, it is necessary to make use of a electromyography sensor. The one chosen for the final version, due to availability and low cost, was a generic EMG sensor similar to the Muscle Sensor V3, by Advancer Technologies [3] shown in Figure 4. This sensor is made to be used with a microcontroller board, such as an Arduino or an ESP32. The signal is also said to be rectified and smoothed, so it can be directly read by the board's ADC converter.

As stated on the user's manual [3], the sensor needs a dual polarity DC power supply (two 9 volt batteries are suggested), with voltage between 3 and 30 volts, 5 volts being typical. To

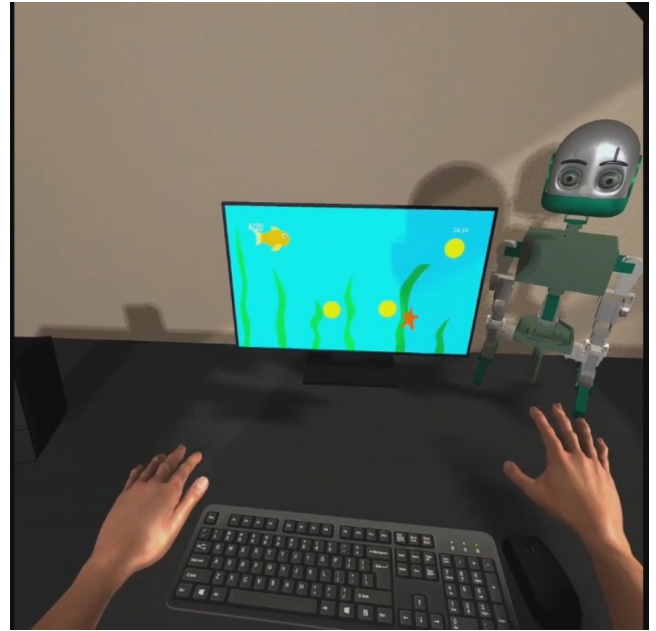


Fig. 3. The game running on a VR environment.



Fig. 4. The EMG sensor used in this project.

avoid a more cumbersome setup and separate batteries, the sensor was connected to a circuit designed to provide dual polarity power just from a microcontroller's DC output pin. The circuit is shown in Figure 5. While the circuit did power the sensor as intended, the signal appears to be afflicted by more noise than before.

V. SIGNAL PROCESSING AND RESULTS

The recorded EMG signal exhibits high frequency noise, with high amplitude variation due to it. In past iterations [2] a

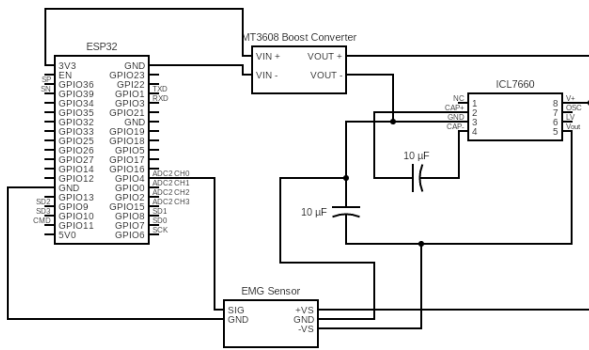


Fig. 5. The circuit responsible for powering the sensor from the microcontroller's output

moving average was implemented, with acceptable results, but increased noise in the latest iteration caused the average values to become lower, suppressing high valued EMG readings, expected when the observed muscle is contracted. Some signal processing algorithms were considered to replace this moving average, they are shown alongside the raw signal in Figure 6.

These algorithms are run on the microcontroller board reading the sensor signal and responsible for transmitting the readings to the game engine via Bluetooth. The tests here presented were run on an Arduino Uno board. An ESP32 was also tested running the same code, and is the board mainly used for the serious game, as it has integrated Bluetooth capabilities.

A simple median filter was tested, as it may reduce noise without affecting peaks as much as an average filter does. The median was calculated with batches of 11 readings, and the resulting signal was far more readable than the original signal, even if the peaks are attenuated to around a third of their original value and delayed. This filter's readings can be seen in red on Figure 6.

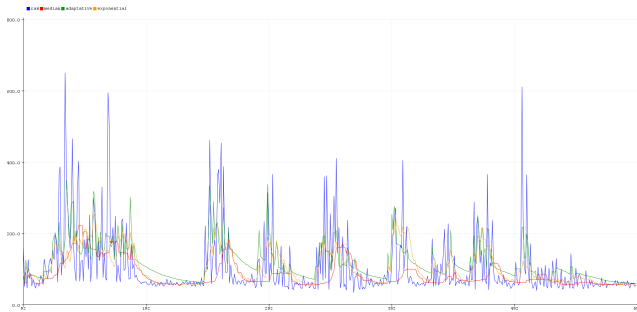


Fig. 6. EMG signal readings, as captured from the sensor and processed by different methods. Blue represents the raw sensor reading, red represents the signal through a median filter, green is a adaptive low-pass filter and yellow an exponential filter. Data captured on the Arduino IDE

A simple low-pass filter could smooth the signal but also attenuate peaks more than desired. In order to reduce this issue, an adaptive low-pass filter was implemented. This filter calculates the variation rate between values and chooses

between two different smoothing coefficients accordingly. This implementation is shown as green on Figure 6, exhibiting the fastest and more noticeable peak response, while filtering out a lot of noise.

Lastly, there's a exponential filter, where the filtered signal is a pondered combination between the current value of the signal and the previously filtered value. This filtered signal is shown as yellow in Figure 6. This method seems to result in the best overall signal, displaying the most clear highs when the muscle was contracted, even if its peaks are far lower when compared to the adaptive filter.

Other approaches were experimented on, but did not provide any meaningful results.

VI. CONCLUSION

While this work provides no definitive answer for which is the absolute best digital filter for EMG signals, it shows some that may be useful depending on the desired application.

For the purpose of this serious game and EMG sensor setup, the exponential filter may be the one with more potential. All of the tested filters resulted in an very attenuated signal that may require some adapting in the game code, but the readings are much clearer than in the non-filtered signal, so the revised code should not face much changes apart from altered thresholds and minor value adjustments. It's also important to note that only the median filter showed any noticeable signal delay.

Even though more testing shall be done to fine tune parameters and achieve better results, these are promising initial results, showing which techniques may be worth more effort.

REFERENCES

- [1] E. C. A. Latorre, G. Valeriano de Barros, G. A. Vieira, G. d. A. Moreira, L. C. Pereira, L. A. Vieira, R. Franco de Oliveira, and R. M. d. Silva, "The effectiveness of gametherapy in the rehabilitation of patients with motor sequelae post-stroke: A systematic review (in Portuguese)," in *Anais da Mostra Acadêmica do Curso de Fisioterapia*, 2020.
- [2] G. R. Silva, and R. S. Guerra, "EMG Based Serious Game For Use in Stroke Rehabilitation." *Anais Estendidos do XXII Simpósio Brasileiro de Jogos e Entretenimento Digital*. SBC, 2023.
- [3] Advancer Technologies (2013). Three-lead differential muscle/electromyography sensor for microcontroller applications. Datasheet
- [4] H. Qassim, W. Z. W. Hasan, H. R. Ramli, H. Harith and L. N. I. Mat (2022). Biceps and triceps electrical activity analysis based on using low-cost sensor: case study. *IEICE Proceedings Series*, 69(SS1-4).
- [5] A. L. Rincon, H. Yamasaki and S. Shimoda, "Design of a video game for rehabilitation using motion capture, EMG analysis and virtual reality," 2016 International Conference on Electronics, Communications and Computers (CONIELECOMP), Cholula, Mexico, 2016, pp. 198-204, doi: 10.1109/CONIELECOMP.2016.7438575. keywords: Sensors;Electromyography;Medical treatment;Engines;Muscles;Games;Three-dimensional displays;virtual reality;muscle rehabilitation;video game;EMG signal;motion capture,
- [6] A Gunadhi, F Z Mustofa, L Agustine, H Pranjoto, R Sitepu and Yuliati (2019, December). Microcontroller-based assistive device for training weak biceps brachii muscle. In *IOP Conference Series: Materials Science and Engineering* (Vol. 673, No. 1, p. 012070). IOP Publishing.

- [7] D. Malathi, P. Naveen, V. Sahasra and S. Rithika, "VLSI Implementation of Median Filter for Noise Removal in EMG Signal," 2023 4th International Conference on Signal Processing and Communication (ICSPC), Coimbatore, India, 2023, pp. 422-426, doi: 10.1109/ICSPC57692.2023.10125903. keywords: Power demand;Working environment noise;Noise reduction;Computer architecture;Signal processing;Muscles;Very large scale integration;EMG;Denoising;Median Filter;Power Consumption;Clock-gating,
- [8] N. Miljković, N. Popović, O. Djordjević, L. Konstantinović, and T. B. Šekara, "ECG artifact cancellation in surface EMG signals by fractional order calculus application," *Computer Methods and Programs in Biomedicine*, vol. 140, pp. 259-264, 2017.
- [9] N. Popovic, N. Miljkovic, O. Djordjevic, and T. B. Sekara, "Artifact cancellation using median filter, moving average filter, and fractional derivatives in biomedical signals," *Proc. of the ICFDA*, vol. 68, no. 2, pp. 150-161, 2016.
- [10] K. Nitta, T. Kasahara, H. Tanaka, H. Kanke, Y. Koyama, M. Toyokura, and Y. Masakado (2019). The clinical efficacy of botulinum toxin injections to the upper lumbrical muscles for clenched fist deformity in chronic stroke patients. *Tokai J Exp Clin Med*, 44(2), 34-39.